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Software Engineering Qualifying Exam

1. What is Reuse-oriented testing?

Ans. Reuse oriented testing is an software testing approach in which existing, reusable software components are tested before integrating them into a new system. It ensures that reused components work correctly, meet requirements, and are compatible with the new system.
Example - Testing an existing login module before using it in a new application.

2. Describe the stages of testing. What are Regression testing and Unit testing?

Ans. Software testing generally consists of 4 stages -
1. Unit testing : Individual functions or modules are tested separately to identify errors.
2. Integration testing : Combined modules are tested to ensure they work together correctly.
3. System testing : The complete software system is tested against it's requirements.
4. Acceptance testing : The system is tested by users or clients to confirm that it meets their expectations.

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Regression Testing: It is performed after software changes or bug fixes to ensure that previously working features are not affected.

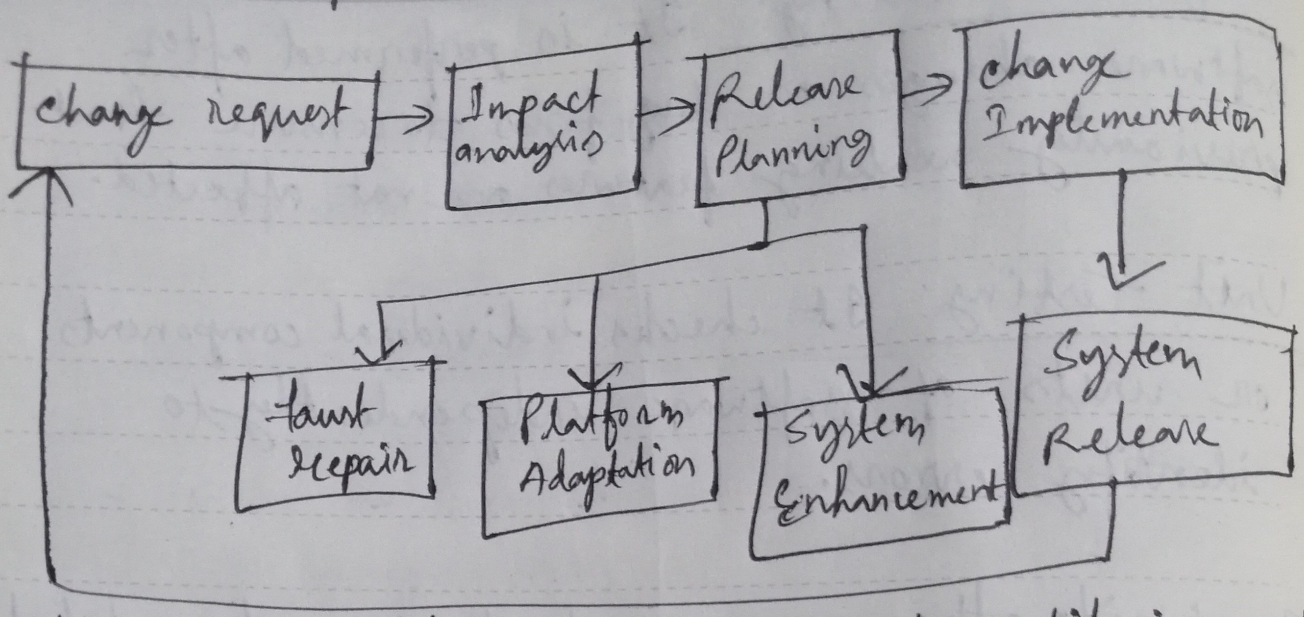
Unit Testing: It checks individual components or units of software independently to identify errors.

3. Write the names and draw and label the steps of the software evolution process.

Ans: The main steps of the Software Evolution Process are -

(I) change identification
↓
(II) change proposal
↓
(III) change Analysis & cost estimation
↓ / or analysis system requirements
(IV) change Implementation
↓
(V) System Release

More Acceptable version:



Software evolution is the process of modifying and updating existing software to meet new requirements, fix bugs, and improve performance.

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4. Describe the process: How to convert an existing system into new system?

Ans: converting an existing system into new system involves the following steps:

(i) System Analysis: Study the existing system and identify its features and limitations.

(ii) Requirement ~~identification~~ identification: Identify the requirements and improvements needed in the new system.

(iii) System Design: Design the architecture, dataleave, and interfaces of the new system.

(iv) Data Migration: Transfer existing data into the new system.

(v) Testing: Test the new system to ensure that it works correctly.

(vi) System Deployment: Install and launch the new system.

(vii) maintanance: Monitor and update the system when necessary.

5. Describe the Software Design Process Steps or Activities.

Ans. The main activities of the Software Design Process are:

1. Architectural Design: Defines the overall structure and components of the software.
2. Database Design: Defines how data will be stored, organized, and managed.
3. Interface Design: Defines how users and different systems components interact.
4. Component Design: Defines the internal structure and functionality of individual components.
5. Algorithm Design: Defines the logical steps required to perform specific operations.

Software design transforms software requirements into a structured blueprint that developers follow during implementation.

